

How Curie works

PRODUCT THESIS

LLMs at the edges. Determinism at the core.

Use models for ambiguous semantic and human-workflow boundaries. Keep scoring, temporal criteria, episode transitions, and interruptive routing versioned and deterministic.

01

Connect

Adaptive onboarding, validation, and exception handling for hospital interfaces.

02

Signal

Deterministic multi-condition surveillance, episode arbitration, and alert governance.

03

Copilot

Grounded explanations, review assistance, protocol retrieval, and feedback analytics.

C4 architecture — LLM workflows

Context → containers → components · LLM stays outside the scoring spine

1 · Context

2 · Containers

3 · Components

LLM boundary

C4 Level 1 — System context. Hospitals send clinical data into Curie; clinicians receive governed alerts and additive explanations. LLMs never sit on the direct path from score to page.

Clinicians / ops

Consumers of pages + narratives

Curie Signal

Deterministic scores · governance · episodes

Curie Connect

FHIR / mapping · trust boundary

EHR / labs / devices

Source systems

Copilot (LLM)

Additive explain · never scores

Static C4 view (CSP-safe) · Caption and legend describe this level

State of view (see table), caption and legend describe this level.

- People interact with alerts and review workbenches — not raw model output as truth.
- External EHR / interface engines remain the system of record.
- Curie is one platform with three product layers: Connect, Signal, Copilot.

Animated LLM workflow portfolio

WF-01 → WF-10 · Play or step each path

WF-01 Mapping

WF-02 Fast

WF-02 Exception

WF-03 Extract

WF-04 Fidelity

WF-05 Review

WF-06 Narrative

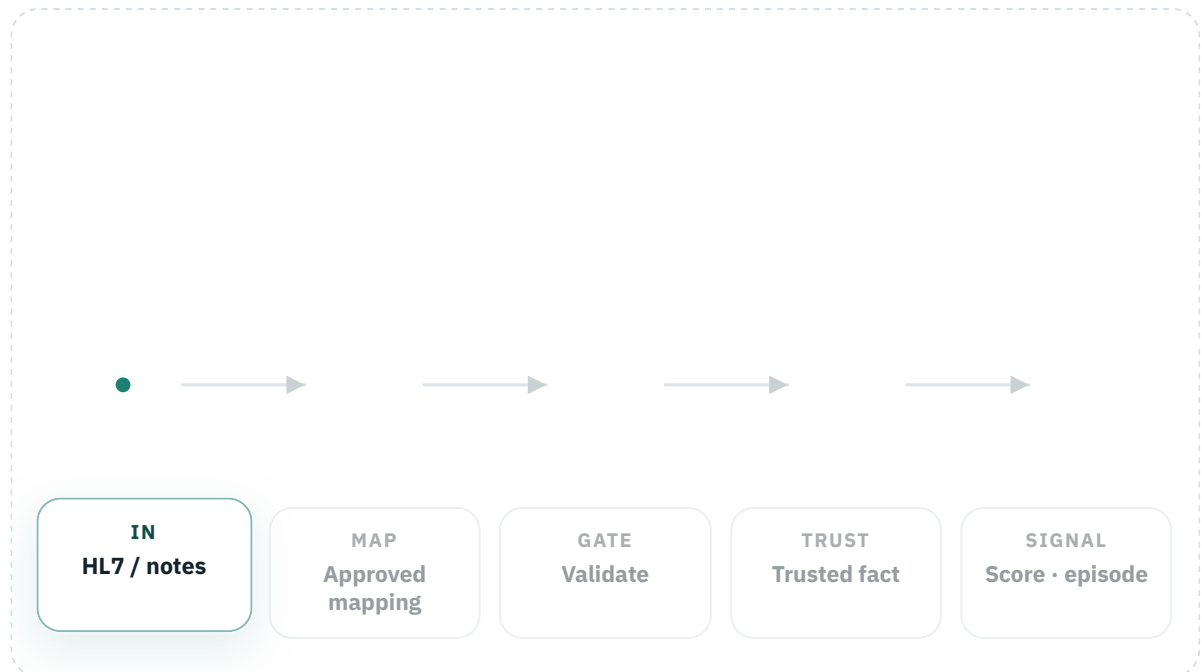
WF-07 Uncertainty

WF-08 Protocol

WF-09 Steward

WF-10 Research

Known message types arrive at Connect's front door.



Play

Step

Reset

Step 1 / 5 · WF-02 · Deterministic fast path

Hard safety boundary

An LLM must not calculate scores, open or route episodes, choose thresholds, invent uncited facts, or promote candidates to trusted state. Model failure never blocks deterministic delivery.

No SOFA / KDIGO / phenotype mutation

No episode open / suppress / escalate

› No live threshold or rule activation

› Every claim cites allowed evidence IDs

LLM workflow portfolio

What runs where · try live demos below

WF-01 Hospital interface onboarding copilot CONNECT · PLANNED

Propose HL7/FHIR mappings; compile to deterministic transforms.

WF-02 Deterministic-first conversion + LLM exceptions CONNECT · PLANNED

Fast path for known traffic; model only on novel or failed inputs.

WF-03 Provenance-preserving note extraction BRIDGE · DEMO

Candidate facts with spans; promote only after validation.

WF-06 Grounded multi-condition episode copilot LIVE IN DEMO

One narrative for the episode; every claim cites evidence IDs.

WF-07 Uncertainty-band context assistant EVAL · PASSIVE

Passive context only when deterministic evidence is borderline.

WF-09 Alert feedback & stewardship copilot LIVE IN DEMO

Classify dismissals into a governance taxonomy; propose offline experiments.

WF-08 Site-approved protocol retrieval PLANNED

Cite curated institutional protocols only — no free-form prescribing.

WF-10 Research cohort & adjudication copilot PLANNED

Draft packets and phenotypes; LLM labels are proposals, not truth.

WF-06 · Episode narrative

Additive copilot on immutable episode context. Score and routing

WF-09 · Stewardship classify

Classify acknowledgement text into a governance taxonomy. Never mutates

stay unchanged.

EPISODE

episode-53d5a356-61a8-319d-9 ↕

Generate episode explanation

live rules.

FEEDBACK TEXT

Already on fluids and antibiotics;
page was appropriate but not
actionable right now.

Classify feedback

Claims matrix

Investor posture · not regulatory · see docs/research/claims-matrix.md

DEMONSTRATED

- DET-STREAM
Deterministic multi-indicator scoring with versioned rule bundles and evidence IDs
- GOV-VOLUME Shared governance reduces interruptive volume vs naive thresholding while preserving detection on offline evals
- EPISODE-ARB Multiple correlated signals aggregate into one patient episode with dominant problem + page arbitration
- CDS-BOUNDARY CDS Hooks / FHIR evidence presentation boundary without rescoring
- OPS-SEC Production-shaped auth/CORS, ops status, and kill switches

UNDER EVALUATION

- RELIABILITY Duplicate delivery, out-of-order events, and process restart do not corrupt alert/episode identity
- MIMIC-STAGE-B
MIMIC-IV Stage B clinical retrospective with locked temporal holdout
- SHADOW-PROD Silent prospective / shadow-mode hospital deployment metrics
- LLM-STEWARD LLM feedback classification improves alert stewardship

NOT CLAIMED

- DX-SEPSIS Diagnoses sepsis (or AKI / respiratory failure) for a patient
- OUTCOME-MORT Improves mortality, organ failure, or time-to-antibiotics
- CLIN-VALID Clinically validated for patient care
- REG-CLEAR FDA cleared / SaMD authorized
- SUPERIOR-NEWS Superior to NEWS/qSOFA/vendor CDS across sites

Investor demo snapshot

Multi-signal → one episode · chaos passed

SIGNALS → EPISODE

NAIVE PAGES

EPISODE PAGES

5 → 1

sofa-deterioration

5

Every emission

1

vs 1 passive

EVIDENCE + HASHES

5

alerts with rule digests

DEMO

Not clinically validated · Not for patient care · Scores are deterministic prototypes